

Joel Oliveira Reis

Assistant Professor, University of Macau

Address

Room E11-4053 FST
University of Macau, E11
Avenida da Universidade
Taipa, Macau, China

Contact

joel.oliveira.reis@gmail.com
+85365317252

Languages

Native Portuguese speaker
Full professional proficiency
in English

Skills

Matlab, Python, C/C++,
ROS, Git, Linux,
SolidWorks, L^AT_EX

ORCID iD

0000-0002-9281-6829

Github

joelreis

CV last updated on August 16, 2026

I work at the intersection of engineering and applied mathematics, transforming theory into practical solutions. My experience includes hands-on work in electrical and mechanical workshop routines, network management, and 3D CAD design, which provided the foundation for my academic research, primarily focused on estimation and control theory. This research has been realized in autonomous platforms ranging from aerial drones to underwater vehicles. Exposure to diverse theoretical and practical systems has equipped me with a broad set of tools and knowledge across multiple engineering domains.

Education

2014 – 2019

Aug – Nov

Doctor of Philosophy in Electrical and Computer Engineering

University of Macau

2011 – 2013

Sep – Nov

Master in Aerospace Engineering

Instituto Superior Técnico, University of Lisbon

2008 – 2011

Sep – Jun

Graduated Engineering Sciences - Aerospace Engineering

Instituto Superior Técnico, University of Lisbon

Work Experience

2024 –

Jan –

Assistant Professor (Research Track)

Faculty of Information Science and Computing, University of Macau

Engaged in student research supervision, securing research funding, and delivering specialized courses such as Marine Robotics and Applications and Ocean Acoustics.

2022 – 2023

Mar – Dec

Post-doctoral Fellow

Faculty of Science and Technology, University of Macau

Period of two years, funded by the FDCT Funding Scheme for Postdoctoral Researchers of Higher Education Institutions of the UM Macao Talent Programme, dedicated to the practical development and testing of prototypes for autonomous aerial and underwater vehicles.

2019 – 2022

Dec – Mar

Research Assistant

Faculty of Science and Technology, University of Macau

Involved in UM Funded projects STEALTH (S^Tate Estim^Ation in Large neTworks with Heterogenous agents) and SECANTS (Self-triggered and Event-triggered Control of Autonomous NeTworked Systems), and in FDCT funded project SLOTMAV (Slung Load Transportation by Multiple Aerial Vehicles).

2012 – 2014

Oct – Aug

Fellow Researcher

Institute for Systems and Robotics, Lisbon

Worked in the Project MAST/AM: Advanced Tracking and Telemetry Methodologies to Study Marine Animals, which was funded by FCT — *Fundação para a Ciência e Tecnologia*. Main work consisted in C-programming a Digital Signal Processor C6713 from Texas Instruments that served as the backbone of an ultra-short baseline acoustic positioning system.

Awards and Recognition ★

2024 **IEEE Journal of Oceanic Engineering's Outstanding Reviewer**

2022 **FDCT Funding Scheme for Postdoctoral Researchers of Higher Education Institutions**

Macau, China

Awarded a post-doctoral fellowship for a period of 24 months.

2020 **First Zhuhai Wanshan International Intelligent Vessel Competition**

Zhuhai, China

Winner of Sailing Race Project - Development of a path following controller for an unmanned surface vessel.

2020 **Macau Science and Technology Development Fund (FDCT)**

Macau, China

Recipient of Scientific and Technological R&D Award for Postgraduates.

Professional Service ✎

- ✎ **Journal Reviewer:** IEEE Transactions on Automation and Science and Engineering, IEEE Transactions on Automatic Control, IEEE Transactions on Cybernetics, IEEE Transactions on Systems, Man, and Cybernetics, IEEE/ASME Transactions on Mechatronics, IEEE Transactions on Aerospace and Electronic Systems, IEEE Aerospace and Electronic Systems Magazine, IEEE Transactions on Cognitive and Developmental Systems, IEEE Sensors Journal, IEEE Transactions on Instrumentation and Measurement, Automatica, Asian Journal of Control, The International Journal of Robotic Research, IEEE Journal of Oceanic Engineering, Ocean Engineering
- ✎ **Conference Reviewer:** IEEE Conference on Decision and Control (CDC), IEEE American Control Conference (ACC), IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), IEEE International Conference on Robotics and Automation (ICRA), The International Conference on Unmanned Aircraft Systems (ICUAS)

Publications 📖

in Journals

- ✎ P. Santos, **J. Reis**, P. Oliveira, and C. Silvestre, "QuadRocket: An Aerial Robotic Testbed for Adaptive Thrust-Vector Control of Rocket-Like Vehicles," in IEEE Transactions on Aerospace and Electronic Systems, vol. xx, no. xx, pp. xx-xx, 2026 (in press).
- ✎ P. Santos, **J. Reis**, P. Oliveira, and C. Silvestre, "Geometric Reduced-Attitude Tracking under a Time-Varying Conic Constraint via Smooth Reference-Shaping," in IEEE Control Systems Letters, vol. xx, no. xx, pp. xx-xx, Oct. 2026.
- ✎ **J. Reis**, and C. Silvestre, "Attitude Estimation on SO(3) Under Biased, Scaled and Misaligned Rate Gyro Measurements," in IEEE Transactions on Automatic Control, vol. xx, no. xx, pp. xx-xx, Oct. 2026.
- ✎ **J. Reis**, and C. Silvestre, "Bottom-following control of AUVs using multibeam forward-looking sonar," Control Engineering Practice, vol. 172, Elsevier BV, p. 106874, Jul. 2026.
- ✎ M. Khodaverdian, **J. Reis**, and C. Silvestre, 'Koopman-based discrete-time controller design for nonlinear flexible spacecraft dynamics', IEEE Trans. Aerosp. Electron. Syst., pp. 1–16, 2026.
- ✎ W. Yang, **J. Reis**, G. Yu, L. Kong and C. Silvestre, "Trajectory Generation and Extended High-Gain Observer-Based Output Feedback Control for an Underactuated Flying Inverted Pendulum," in IEEE Transactions on Automation Science and Engineering, vol. 23, pp. 2440-2453, 2026

- ✍ W. Yang, **J. Reis**, G. Yu, and C. Silvestre, "Output feedback control of an underactuated flying inverted pendulum", *Control Engineering Practice*, vol. 164, Elsevier BV, p. 106474, Nov. 2025.
- ✍ **J. Reis**, and C. Silvestre, "Kinematics-informed neural network control on $SO(3)$ ", *Automatica*, vol. 174, Elsevier BV, p. 112172, Apr. 2025.
- ✍ L. Kong, **J. Reis**, W. He and C. Silvestre, "Hybrid Triggering Adaptive Prescribed Performance Control of Underactuated UAVs," in *IEEE Transactions on Industrial Electronics*, vol. 73, no. 7, pp. 10242-10253, July 2026
- ✍ L. Kong, **J. Reis**, W. He and C. Silvestre, "Event-triggered prescribed-time tracking control for UAVs using polynomial error trajectories," in *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 16476-16486, May 2025
- ✍ Y. Ju, **J. Reis**, Y. Sun and C. Silvestre, "On the Asymptotic Stability of 2-D Linear Inhomogeneous Systems," in *IEEE Transactions on Automatic Control*, vol. 70, no. 8, pp. 5546-5552, Aug. 2025.
- ✍ W. Yang, G. Yu, **J. Reis**, and C. Silvestre, "Robust nonlinear 3D control of an inverted pendulum balanced on a quadrotor," *Automatica*, vol. 159, Elsevier BV, p. 111336, Jan. 2024.
- ✍ L. Kong, **J. Reis**, W. He, X. Yu, and C. Silvestre, "On dynamic performance control for a quadrotor-slung-load system with unknown load mass," *Automatica*, vol. 162, Elsevier BV, p. 111516, Apr. 2024.
- ✍ G. Yu, **J. Reis**, D. Cabecinhas, R. Cunha, and C. Silvestre, "Reduced-Complexity Active Disturbance Rejection Controller for Quadrotor-Slung-Load Transportation," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 53, no. 8. Institute of Electrical and Electronics Engineers (IEEE), pp. 5248–5259, Aug. 2023.
- ✍ G. Yu, **J. Reis**, and C. Silvestre, "Quadrotor Neural Network Adaptive Control: Design and Experimental Validation," *IEEE Robotics and Automation Letters*, vol. 8, no. 5. Institute of Electrical and Electronics Engineers (IEEE), pp. 2574–2581, May 2023.
- ✍ **J. Reis**, W. Xie, D. Cabecinhas, and C. Silvestre, "Nonlinear Backstepping Controller for an Underactuated ASV With Model Parametric Uncertainty: Design and Experimental Validation," *IEEE Transactions on Intelligent Vehicles*, vol. 8, no. 3. Institute of Electrical and Electronics Engineers (IEEE), pp. 2514–2526, Mar. 2023.
- ✍ **J. Reis**, P. Batista, P. Oliveira, and C. Silvestre, "Discrete-time Kalman filter for heave motion estimation," *Ocean Engineering*, vol. 277. Elsevier BV, p. 114240, Jun. 2023.
- ✍ **J. Reis**, G. Yu, and C. Silvestre, "Kalman-based velocity-free trajectory tracking control of an underactuated aerial vehicle with unknown system dynamics," *Automatica*, vol. 155. Elsevier BV, p. 111148, Sep. 2023.
- ✍ L. Kong, **J. Reis**, W. He, and C. Silvestre, 'Experimental validation of a robust prescribed performance nonlinear controller for an unmanned aerial vehicle with unknown mass', *IEEE ASME Trans. Mechatron.*, vol. 29, no. 1, pp. 301–312, Feb. 2024.
- ✍ L. Kong, **J. Reis**, W. He, and C. Silvestre, "Comprehensive Nonlinear Control Strategy for VTOL-UAVs With Windowed Output Constraints," *IEEE Transactions on Control Systems Technology*, vol. 31, no. 6. Institute of Electrical and Electronics Engineers (IEEE), pp. 2673–2684, Nov. 2023.
- ✍ **J. Reis**, P. Batista, P. Oliveira, and C. Silvestre, "Earth Velocity and Rigid-Body Attitude Estimation on $SO(3)$ Using Biased Measurements," *IEEE/ASME Transactions on Mechatronics*, vol. 27, no. 6. Institute of Electrical and Electronics Engineers (IEEE), pp. 4246–4257, Dec. 2022.
- ✍ **J. Reis**, G. Yu, D. Cabecinhas, and C. Silvestre, "High-performance quadrotor slung load transportation with damped oscillations," *International Journal of Robust and Nonlinear Control*, vol. 33, no. 17. Wiley, pp. 10227–10256, Aug. 04, 2022.
- ✍ **J. Reis**, P. Batista, P. Oliveira, and C. Silvestre, "Attitude, body-fixed Earth rotation rate, and sensor bias estimation using single observations of direction of gravitational field," *Automatica*, vol. 125. Elsevier BV, p. 109475, Mar. 2021.
- ✍ W. Xie, **J. Reis**, D. Cabecinhas, and C. Silvestre, "Design and experimental validation of a nonlinear controller for underactuated surface vessels," *Nonlinear Dynamics*, vol. 102, no. 4. Springer Science and Business Media LLC, pp. 2563–2581, Nov. 06, 2020.

- ✎ **J. Reis**, P. Batista, P. Oliveira, and C. Silvestre, "Kalman Filter Cascade for Attitude Estimation on Rotating Earth," *IEEE/ASME Transactions on Mechatronics*, vol. 25, no. 1. Institute of Electrical and Electronics Engineers (IEEE), pp. 327–338, Feb. 2020.
- ✎ **J. Reis**, P. Batista, P. Oliveira, and C. Silvestre, "Calibration of High-Grade Inertial Measurement Units Using a Rate Table," *IEEE Sensors Letters*, vol. 3, no. 4. Institute of Electrical and Electronics Engineers (IEEE), pp. 1–4, Apr. 2019.
- ✎ **J. Reis**, P. Batista, P. Oliveira, and C. Silvestre, "Nonlinear Observer on $SO(3)$ for Attitude Estimation on Rotating Earth Using Single Vector Measurements," *IEEE Control Systems Letters*, vol. 3, no. 2. Institute of Electrical and Electronics Engineers (IEEE), pp. 392–397, Apr. 2019.
- ✎ **J. Reis**, P. Batista, P. Oliveira, and C. Silvestre, "Attitude estimation using high-grade gyroscopes," *Control Engineering Practice*, vol. 92, Elsevier BV, p. 104134, Nov. 2019.
- ✎ **J. Reis**, P. Batista, P. Oliveira, and C. Silvestre, "Source Localization Based on Acoustic Single Direction Measurements," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 54, no. 6. Institute of Electrical and Electronics Engineers (IEEE), pp. 2837–2852, Dec. 2018.
- ✎ **J. Reis**, M. Morgado, P. Batista, P. Oliveira, and C. Silvestre, "Design and Experimental Validation of a USBL Underwater Acoustic Positioning System," *Sensors*, vol. 16, no. 9. MDPI AG, p. 1491, Sep. 14, 2016.

in Conferences

- ✎ **J. Reis**, and C. Silvestre, "Bottom-following control of AUVs using multibeam forward-looking sonar," in proceedings of IFAC World Congress 2026, Aug. 2026.
- ✎ **J. Reis** and C. Silvestre, 'Kalman filter as observer and smoother for rigid-body motion control applications', *IFAC-PapersOnLine*, vol. 58, no. 16, pp. 95–100, 2024.
- ✎ P. Casau, **J. Reis**, and C. Silvestre, "Event-triggered trajectory tracking control of an underactuated autonomous surface vessel," 2022 American Control Conference (ACC). IEEE, Jun. 08, 2022.
- ✎ **J. Reis**, C. Silvestre, P. Batista, and P. Oliveira, "Attitude observers aided by implicit measurements of the Earth angular velocity," 2020 59th IEEE Conference on Decision and Control (CDC). IEEE, Dec. 14, 2020.
- ✎ **J. Reis**, P. Batista, P. Oliveira, and C. Silvestre, "Nonlinear Attitude Observer on $SO(3)$ Based on Single Body-Vector Measurements," 2018 IEEE Conference on Control Technology and Applications (CCTA). IEEE, Aug. 2018.
- ✎ **J. Reis**, P. Oliveira, P. Batista, and C. Silvestre, "Filter design for localization aided by direction and Doppler measurements," 2014 IEEE International Conference on Robotics and Automation (ICRA). IEEE, May 2014.